

EE/CS 148B - Homework 3

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June 1, 2026

Vision Transformer

- (a) `vit_pooling`: For tasks requiring spatial reasoning, attention pooling is expected to perform best because it can learn to weight tokens selectively based on task-relevant spatial cues, rather than treating all positions equally (mean pooling) or relying on a single generic summary (CLS). Mean pooling discards per-patch identity by averaging, which collapses distinctions between, say, a red cube on the left versus the right; the CLS token suffers the same problem more acutely, since it is trained to summarize global semantics and does not preserve fine-grained spatial layout at all. When an entire image is condensed into a single CLS vector before being passed to a language model, all explicit positional relationships between objects are lost. The model receives no direct signal about where each object is, only a holistic blend of what is in the image.
- (b) `vit_patch_size`:
- (1) Patch counts and attention cost.** For a 224×224 image the number of patches is $N = (224/P)^2$:

P	N
8	784
16	196
32	49

Self-attention scales as $O(N^2 d_{\text{model}})$, and $N \propto P^{-2}$. Thus, the overall cost scales as $O(P^{-4} d_{\text{model}})$. Halving the patch size increases attention compute by $2^4 = 16\times$, so shrinking P is extremely expensive.

(2) Wall-clock times (batch of 16 images, $d_{\text{model}} = 384$, 6 heads, 6 blocks; RTX 4070 Laptop GPU; 5 warmup + 20 timed steps):

P	Mean (ms)	Std (ms)
8	84.9	1.8
16	14.8	1.3
32	10.1	1.3

The empirical ratios ($8.4\times$ between $P = 8$ and $P = 16$, and $1.5\times$ between $P = 16$ and $P = 32$) are smaller than the theoretical $16\times$ and $256\times$ because the GPU is compute-bound only for the largest sequence length; at $P = 16$ and $P = 32$ the bottleneck shifts to memory bandwidth and kernel launch overhead, compressing the observed differences. The $P = 8$ overhead, roughly $8\times$ slower than $P = 16$, confirms that the quadratic attention term dominates at high patch counts.

(3) Accept smaller patches when fine-grained spatial fidelity is critical and the compute budget allows it (e.g., OCR, dense prediction, or counting small objects where coarser tokens would merge distinct features).

CLIP-Style Contrastive Pretraining

- (a) infonce: The loss is averaged in both directions because each (image, text) pair should serve as a positive for both retrieval directions—finding the right caption given an image, and the right image given a caption. Using only one direction would bias the model toward one retrieval mode and leave half the pairwise supervision signal in each batch unused.